Lab Quetsion

1.Design a Java program for an ATM where an exception is generated when the user attempts to withdraw an amount greater than the available balance.

Program:

```
class InsufficientBalanceException extends Exception {

    InsufficientBalanceException(String message) {
        super(message);
    }
}

class ATM {
    double balance;

    ATM(double balance) {
        this.balance = balance;
    }

    void withdraw(double amount) throws InsufficientBalanceException {
        if (amount > balance) {
            throw new InsufficientBalanceException(
                "Insufficient Balance! Withdrawal amount is greater than available balance."
            );
        }

        balance = balance - amount;
        System.out.println("Withdrawal Successful.");
        System.out.println("Withdrawn Amount : " + amount);
        System.out.println("Remaining Balance: " + balance);
    }
}

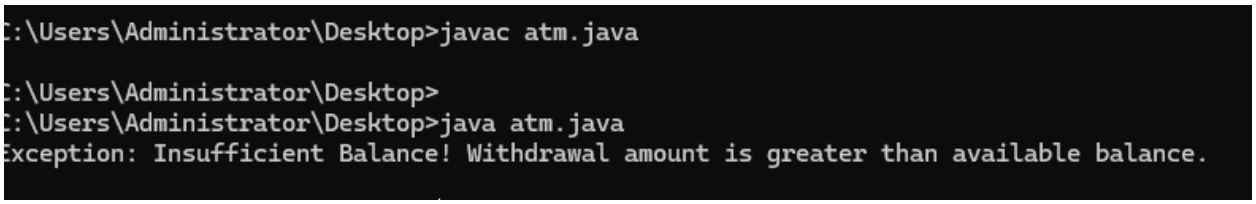
public class atm {
    public static void main(String[] args) {

        ATM atm = new ATM(5000);

        try {
            atm.withdraw(7000);
```

```
        }
        catch (InsufficientBalanceException e) {
            System.out.println("Exception: " + e.getMessage());
        }
    }
}
```

Output:



2.Create a user-defined exception called InvalidAgeException for an online voting system. The exception should occur when the age is below 18.

Program:

```
class InvalidAgeException extends Exception {

    InvalidAgeException(String message) {
        super(message);
    }
}

public class vote {

    static void checkAge(int age) throws InvalidAgeException {

        if (age < 18) {
            throw new InvalidAgeException(
                "You are not eligible to vote. Age must be 18 or above."
            );
        }

        System.out.println("You are eligible to vote.");
    }
```

```
    public static void main(String[] args) {

        int age = 16;

        try {
            checkAge(age);
        }
        catch (InvalidAgeException e) {
            System.out.println("Exception: " + e.getMessage());
        }
    }
}
```

Output:

```
C:\Users\Administrator\Desktop>javac vote.java

C:\Users\Administrator\Desktop>java vote.java
Exception: You are not eligible to vote. Age must be 18 or above.
```

3.Create an interface Payment with a method pay(). Implement it using classes UPIPayment and CardPayment. Explain how interfaces support abstraction.

Program:

```
interface Payment {
    void pay();
}

class UPIPayment implements Payment {

    public void pay() {
        System.out.println("Payment made using UPI.");
    }
}

class CardPayment implements Payment {

    public void pay() {
        System.out.println("Payment made using Credit/Debit Card.");
```

```
    }
}

public class payment {
    public static void main(String[] args) {

        Payment payment1 = new UPIPayment();
        payment1.pay();

        Payment payment2 = new CardPayment();
        payment2.pay();
    }
}
```

Output:

